Example 6.17 Area of the triangle formed by a line with the coordinate axes, is 36 square units. Find the equation of the line if the perpendicular drawn from the origin to the line makes an angle of 45° with positive the x-axis.

Solution:

Let p be the length of the perpendicular drawn from the origin to the required line.
The perpendicular makes 45° with the x-axis.
The equation of the required line is of the form,

$$\begin{aligned} x\cos\alpha + y\sin\alpha &= p \\ \Rightarrow x\cos 45^\circ + y\sin 45^\circ &= p \\ \Rightarrow x + y &= \sqrt{2}p \end{aligned}$$

This equation cuts the coordinate axes at $A(\sqrt{2}p, 0)$ and $B(0, \sqrt{2}p)$.
Area of the ΔOAB is

$$\frac{1}{2} \times \sqrt{2}p \times \sqrt{2}p = 36 \quad \Rightarrow p = 6 \quad (\because \ p \text{ is positive})$$

Therefore the equation of the required line is

$$x + y = 6\sqrt{2}$$

Example 6.18 Find the equation of the lines make an angle 60° with positive x-axis and at a distance $5\sqrt{2}$ units measured from the point $(4, 7)$, along the line $x - y + 3 = 0$.

Solution:

The angle of inclination of the line $x - y + 3 = 0$ is 45°, and a point on the line is $(4, 7)$.
Using parametric form

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r,$$

the above equation can be written as

$$\frac{x-4}{\frac{1}{\sqrt{2}}} = \frac{y-7}{\frac{1}{\sqrt{2}}} = \pm 5\sqrt{2} \text{ (for either side of (4,7) at a distance } r = \pm 5\sqrt{2})$$

That is $x - 4 = y - 7 = \pm 5$.
The points on the lines at a distance $5\sqrt{2}$ units either side of $(4, 7)$ are
$(4 + 5, 7 + 5)$ and $(4 - 5, 7 - 5)$.
The points on the lines are $(9, 12)$ and $(-1, 2)$ and the given slope is $m = \tan 60^\circ = \sqrt{3}$.
Therefore the required equations, using slope and a point form, we get

$$\begin{aligned} \sqrt{3}x - y + (12 - 9\sqrt{3}) &= 0 \text{ and} \\ \sqrt{3}x - y + (2 + \sqrt{3}) &= 0 \end{aligned}$$

6.3.4 General form to other forms

$Ax + By + C = 0$, where A, B and C being real numbers and A and B cannot be simultaneously equal to zero, can be expressed in terms of A, B and C of the general form into other forms.